

$$x_1 = a_1 + 0.01a_2, \quad a_1 = x_1 - 0.01x_2, \quad x_2 = a_2, \quad x_3 = a_3.$$

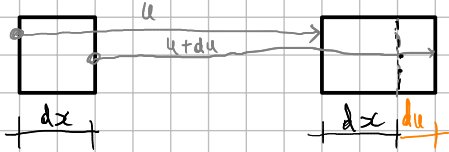
Tenues de petites déformations

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x}, \quad \epsilon_{yy} = \frac{\partial u_y}{\partial y}, \quad \epsilon_{zz} \dots$$

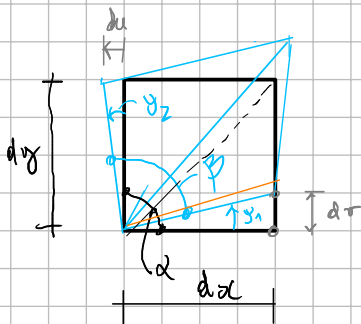
$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right), \quad \epsilon_{yz} = \frac{1}{2} \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right),$$

$$\epsilon_{xz} = \dots$$



$$\epsilon_{xx} = \frac{du}{dx}$$

$$u = u_x \quad v = u_y$$



$$\alpha = \pi/2$$

$$\beta = \frac{\pi}{2} - \gamma_1 + \gamma_2$$

$$\left. \begin{array}{l} \alpha = \pi/2 \\ \beta = \frac{\pi}{2} - \gamma_1 + \gamma_2 \end{array} \right\} \begin{array}{l} \text{unido el ángulo recto de deformación} \\ \alpha - \beta = \gamma_1 - \gamma_2 \end{array}$$

$$\tan(\gamma_1) = \frac{dv}{dx} \approx \gamma_1$$

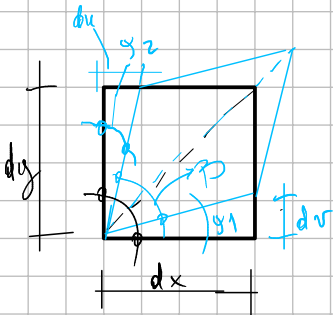
$$\tan(\gamma_2) = -\frac{du}{dy} \approx \gamma_2$$

$$\gamma = \pi/2 / 2$$

$$\Gamma = \frac{\gamma_1 + \pi/2 + \gamma_2}{2}$$

$$\Gamma - \gamma = \frac{1}{2} (\gamma_1 + \gamma_2) = \frac{1}{2} \omega_{xy}$$

$$\alpha - \beta = \gamma_1 - \gamma_2 = \frac{dv}{dx} - \left(-\frac{du}{dy} \right) = \frac{dv}{dx} + \frac{du}{dy} = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} = 2\epsilon_{xy}$$



$$\alpha = \pi/2$$

$$\beta = \pi/2 - \gamma_1 - \gamma_2$$

$$\left. \begin{array}{l} \alpha = \pi/2 \\ \beta = \pi/2 - \gamma_1 - \gamma_2 \end{array} \right\} \alpha - \beta = \gamma_1 + \gamma_2 = \frac{dv}{dx} + \frac{du}{dy} = 2\epsilon_{xy}$$

$$\gamma = \gamma_1 = \gamma_2 \quad (\text{en matrices diagonales})$$

$$du_i = \frac{\partial u_i}{\partial x_j} dx_j$$

$$du = \nabla u \cdot d\mathbf{x}$$

differential $u(x_1, x_2, x_3)$

$$\begin{aligned} du_i &= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} dx_j + \frac{\partial u_i}{\partial x_j} dx_j \right) = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_i}{\partial x_j} \right) dx_j \\ &= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right) dx_j \\ &= \left[\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right) \right] dx_j \end{aligned}$$

ϵ_{ij}

$-w_{ij}$

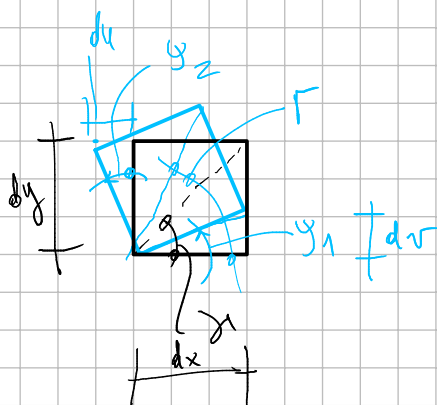
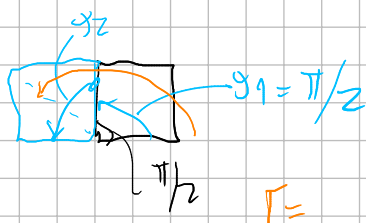
$\text{rot} u = \nabla \times u$

$$\nabla \times v = 0$$

$$(\nabla \times u)_i = \epsilon_{ijk} \frac{\partial u_k}{\partial x_j}$$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_x & u_y & u_z \end{vmatrix} = \left(\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z} \right) i + \dots$$

$$\left(\frac{\partial u_x}{\partial z} - \frac{\partial u_z}{\partial x} \right) j + \dots \cdot k$$



$$\delta = \alpha/2 = \pi/2/2$$

$$\Gamma = \frac{\gamma_1 + (\gamma_2 + \pi/2)}{2}$$

$$\omega = \Gamma - \delta = \frac{1}{2} (\gamma_1 + \gamma_2 + \pi/2 - \pi/2)$$

$$= \frac{1}{2} (\gamma_1 + \gamma_2) = \frac{1}{2} \left(\frac{dr}{dx} - \frac{du}{dy} \right)$$

$$= \frac{1}{2} \left(\frac{du_y}{dx} - \frac{du_x}{dy} \right) = w_{xy}$$

$$x_1 = a_1 + 0.01a_2, \quad a_1 = x_1 - 0.01x_2, \quad \overbrace{x_2 = a_2}, \quad x_3 = a_3.$$

$$u_i = x_i - z_i \quad \Rightarrow \quad \underline{u} = \underline{x} - \underline{z}$$

$$u_1 = x_1 - z_1 = \cancel{z_1} + 0.01z_2 - \cancel{z_1} = 0.01z_2 = u_1$$

$$u_2 = x_2 - z_2 = 0$$

$$u_3 = x_3 - z_3 = 0$$

$$\varepsilon_{xx} = \frac{1}{2} \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_x}{\partial x} \right) = \frac{1}{2} \left(\frac{\partial u_1}{\partial a_1} + \frac{\partial u_1}{\partial a_1} \right) = \frac{\partial u_1}{\partial a_1} = 0$$

$$\varepsilon_{yy} = \frac{\partial u_y}{\partial y} = \frac{\partial u_z}{\partial a_2} = 0$$

$$\varepsilon_{zz} = \frac{\partial u_z}{\partial a_3} = 0$$

$$\varepsilon_{xy} = \frac{1}{2} \left(\frac{\partial u_1}{\partial a_2} + \frac{\partial u_z}{\partial a_1} \right) = \frac{1}{2} (0.01 + 0) = 0.005$$

$$\varepsilon_{xz} = 0$$

$$\varepsilon_{yz} = 0$$